United States Patent	12402261
Kind Code	B2
Date of Patent	August 26, 2025
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Button module

Abstract

A button module, including a bracket, at least one button, at least one first positioning member, and at least one elastic member, is provided. The bracket has a first surface and a second surface connected to the first surface. The button is disposed on the first surface. The first positioning member includes an elastic arm and a hook connected to the elastic arm. The elastic arm is connected to the second surface, and the hook is engaged with a casing. The elastic member protrudes from the first surface and abuts against the casing.

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Appl. No.: **18/447216**

Filed: **August 09, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20240081006 A1	Mar. 07, 2024

Foreign Application Priority Data

TW	111133555	Sep. 05, 2022

Publication Classification

Int. Cl.: **H05K5/02** (20060101)

U.S. Cl.:

CPC **H05K5/0217** (20130101);

Field of Classification Search

CPC: H01H (2003/00); H01H (2003/007); H01H (2003/02); H01H (2003/12); H01H (2009/02); H01H (2013/00); H01H (2013/02); H01H (2013/04); H01H (2013/50); H01H (2013/52); H01H (2223/04); H01H (9/02); H01H (9/0214); H01H (9/0242); H01H (9/04); H01H (9/20); H01H (9/22); H01H (9/28); H01H (13/00); H01H (13/02); H01H (13/04); H01H (13/12); H01H (13/14); H01H (13/20); H01H (13/26); H01H (13/50); H01H (13/52); H01H (3/00); H01H (3/02); H01H (3/12); H01H (3/32); H05K (5/00); H05K (5/02); H05K (5/0217)

USPC: 200/341

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the priority benefit of Taiwan application serial no. 111133555, filed on Sep. 5, 2022. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technology Field

(2) The disclosure relates to a button module, and in particular to a button module applicable to an electronic device.

Description of Related Art

(3) Common electronic devices such as smart phones or tablet computers are mostly provided with power buttons and volume buttons on the sides of the casings, so that users can switch the electronic devices on or off or adjust the volume. Specifically, the side of the casing has at least two openings for accommodating the power button and the volume button, and the power button and the volume button are installed on a bracket to be fixed to the casing through the bracket.

(4) Generally speaking, the bracket is attached to the casing through screws, which not only takes time to disassemble, but also makes it difficult to save manufacturing costs. In addition, during the

process of attaching the screws to the casing, it is easy to cause the screws to be stripped or chipped due to excessive force, and even cause the bracket or the casing to break, resulting in poor assembly yield. In addition, when the user presses the power button or the volume button, the force-bearing bracket may shake due to the tolerance between the screws and screw holes, thereby affecting the operating reliability and the operating texture.

SUMMARY

(5) The disclosure provides a button module, which not only helps to reduce the assembly working hours, but also helps to improve the operating reliability.

(6) The disclosure provides a button module, which includes a bracket, at least one button, at least one first positioning member, and at least one elastic member. The bracket has a first surface and a second surface connected to the first surface. The button is disposed on the first surface. The first positioning member includes an elastic arm and a hook connected to the elastic arm. The elastic arm is connected to the second surface, and the hook is engaged with a casing. The elastic member protrudes from the first surface and abuts against the casing.

(7) In an embodiment of the disclosure, a quantity of the first positioning members is two, and the elastic member is located between the two first positioning members.

(8) In an embodiment of the disclosure, a quantity of the elastic elements is two, and the elastic elements are arranged side by side between the two first positioning members.

(9) In an embodiment of the disclosure, the second surface has at least one groove. The elastic arm extends outward from the groove and protrudes from the first surface.

(10) In an embodiment of the disclosure, the hook of the first positioning member protrudes from an end of the elastic arm and is located outside the groove.

(11) In an embodiment of the disclosure, the button module further includes at least one second positioning member. The bracket further has a third surface opposite the second surface, and the first surface is connected to the second surface and the third surface. The second positioning member includes an elastic arm and a hook connected to the elastic arm. The elastic arm is connected to the third surface, and the hook is engaged with the casing.

(12) In an embodiment of the disclosure, the third surface has at least one groove, the elastic arm of the second positioning member is located in the groove, and the hook of the second positioning member protrudes from the third surface.

(13) In an embodiment of the disclosure, the casing has a first positioning surface, a second positioning surface opposite the first positioning surface, at least one first slot located on the first positioning surface, and at least one second slot located on the second positioning surface. The second surface of the bracket faces the first positioning surface, and the third surface of the bracket faces the second positioning surface. The hook of the first positioning member is engaged with the first slot. The hook of the first positioning member has a stopping surface abutting against an inner wall surface of the first slot. The hook of the second positioning member is engaged with the second slot. The hook of the second positioning member has a stopping surface abutting against an inner wall surface of the second slot.

(14) In an embodiment of the disclosure, the second positioning member further includes an operating portion connected to the elastic arm of the at least one second positioning member. The bracket further has a fourth surface opposite the first surface, and the operating portion protrudes from the fourth surface.

(15) In an embodiment of the disclosure, the elastic member includes a semicircular elastic arm, a semilunar elastic arm, or an arced elastic arm.

(16) Based on the above, in the button module of the disclosure, the positioning member and the elastic member are integrated on the bracket, and the bracket may be fixed to the casing through the positioning member, which is not only convenient for assembly, but is also less prone to breakage of the bracket or casing due to assembly errors, so that assembly working hours can be reduced and assembly yield can be improved. On the other hand, the bracket may abut against the casing

through the elastic member. When the user presses the button, the elastic member may feed back to the user a relatively solid operating texture. Moreover, based on the cooperation between the positioning member and the elastic member, the force-bearing bracket may not easily shake to reduce the idle stroke generated when pressing the button, thereby improving the operating reliability.
(17) In order for the features and advantages of the disclosure to be more comprehensible, the following specific embodiments are described in detail in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** and FIG. **2** are schematic views of a button module in two different viewing angles according to an embodiment of the disclosure.
(2) FIG. **3** is a schematic view of a button module installed on a casing according to an embodiment of the disclosure.
(3) FIG. **4** is a schematic cross-sectional view of FIG. **3** along a section line A-A.
(4) FIG. **5** is a schematic cross-sectional view of FIG. **3** along a section line B-B.
DESCRIPTION OF THE EMBODIMENTS
(5) FIG. **1** and FIG. **2** are schematic views of a button module in two different viewing angles according to an embodiment of the disclosure. FIG. **3** is a schematic view of a button module installed on a casing according to an embodiment of the disclosure. Please refer to FIG. **1** to FIG. **3**. In this embodiment, a button module **100** is applicable to be installed on a casing **10** and is fixed by adopting a screwless manner, which helps to save manufacturing costs. Specifically, the button module **100** includes a bracket **110**, a first button **120***a*, a second button **120***b*, two first positioning members **130**, and two elastic members **140**, wherein the first button **120***a* and the second button **120***b* are arranged side by side and are installed on the bracket **110**. As shown in FIG. **3**, the side of the casing **10** has a first opening **11** and a second opening **12** arranged side by side, wherein the first opening **11** is configured to accommodate the first button **120***a*, and the second opening **12** is configured to accommodate the second button **120***b*. For example, the first button **120***a* may be a power button, and the second button **120***b* may be a volume button.
(6) FIG. **4** is a schematic cross-sectional view of FIG. **3** along a section line A-A, and the section line A-A passes through one of the two first positioning members **130**. FIG. **5** is a schematic cross-sectional view of FIG. **3** along a section line B-B, and the section line B-B passes through one of the two elastic members **140**. As shown in FIG. **1**, FIG. **4**, and FIG. **5**, the two first positioning members **130** and the two elastic members **140** are connected to the bracket **110**, and the bracket **110**, the two first positioning members **130**, and the two elastic members **140** are an integrally formed structure, which is, for example, manufactured by plastic injection molding technology. Specifically, the bracket **110** has a first surface **111** and a second surface **112** connected to the first surface **111**, wherein the first button **120***a* and the second button **120***b* are disposed on the first surface **111**, and the two first positioning members **130** are connected to the second surface **112**. In addition, the two elastic members **140** protrude from the first surface **111** and are arranged side by side between the two first positioning members **130**. As shown in FIG. **3** to FIG. **5**, when the button module **100** is installed on the casing **10**, the bracket **110** is engaged with and fixed to the casing **10** through the two first positioning members **130**, and abuts against the casing **10** through the two elastic members **140**.
(7) As shown in FIG. **1**, FIG. **2**, and FIG. **4**, the button module **100** further includes two second positioning members **150** connected to the bracket **110**. The bracket **110**, the two first positioning members **130**, the two elastic members **140**, and the two second positioning members **150** are an integrally formed structure, which is, for example, manufactured by plastic injection molding

technology. Furthermore, the bracket **110** further has a third surface **113** opposite the second surface **112**, and the first surface **111** is connected to the second surface **112** and the third surface **113**. The two second positioning members **150** are connected to the third surface **113**, and the bracket **110** is engaged with and fixed to the casing **10** through the two first positioning members **130** and the two second positioning members **150**. Since the bracket **110** and the casing **10** are fixed by adopting a screwless manner, manufacturing costs can be saved. In addition, it should be noted that the section line A-A passes through one of the two second positioning members **150**.
(8) As shown in FIG. **1**, FIG. **2**, and FIG. **4**, the casing **10** has a first positioning surface **13**, a second positioning surface **14** opposite the first positioning surface **13**, two first slots **15** located on the first positioning surface **13**, and two second slots **16** located on the second positioning surface **14**. The second surface **112** of the bracket **110** faces the first positioning surface **13** of the casing **10**, and the third surface **113** faces the second positioning surface **14**. The two first positioning members **130** abut against the first positioning surface **13** and are engaged with the two first slots **15**. In contrast, the two second positioning members **150** abut against the second positioning surface **14** and are engaged with the two second slots **16**.
(9) As shown in FIG. **3** and FIG. **4**, the bracket **110** is clamped and positioned between the first positioning surface **13** and the second positioning surface **14** of the casing **10**. When the user presses the first button **120***a* or the second button **120***b*, the force-bearing bracket **110** may not easily shake in a first direction D**1**, and therefore the idle stroke generated when the first button **120***a* or the second button **120***b* are pressed is mitigated, thereby improving the operating reliability. In the embodiment of the present invention, the first direction D**1** is perpendicular to the first positioning surface **13**, the second positioning surface **14**, the second surface **112**, or the third surface **113**.
(10) As shown in FIG. **1** to FIG. **4**, the bracket **110** is engaged with and fixed to the casing **10** through the two first positioning members **130** and the two second positioning members **150**, which is not only convenient for assembly, but also less prone to breakage of the bracket **110** or the casing **10** due to assembly mistakes, so that assembly time can be reduced and assembly yield can be improved. As shown in FIG. **1**, FIG. **3**, and FIG. **5**, the bracket **110** abuts against the casing **10** through the two elastic members **140**. When the user presses the first button **120***a* or the second button **120***b*, the two elastic members **140** may feed back to the user a relatively solid operating texture.
(11) As shown in FIG. **3** to FIG. **5**, based on the cooperation between the first positioning member **130** and the elastic member **140** and the cooperation between the second positioning member **150** and the elastic member **140**, the force-bearing bracket **110** may not easily shake in a second direction D**2** perpendicular to the first direction D**1** (or parallel to the pressing direction of the button), so the idle stroke generated when the first button **120***a* or the second button **120***b* is pressed is mitigated, thereby improving the operating reliability.
(12) As shown in FIG. **1** and FIG. **4**, in this embodiment, each first positioning member **130** includes an elastic arm **131** and a hook **132**. The elastic arm **131** is connected to the second surface **112**, and the hook **132** is connected to the elastic arm **131**. In detail, the second surface **112** has two grooves **112***a* configured to accommodate the two elastic arms **131** of the two first positioning members **130**. The two elastic arms **131** extend outward from the two grooves **112***a* and protrude from the first surface **111**. In addition, each hook **132** protrudes from an end of the corresponding elastic arm **131** and is located outside the corresponding groove **112***a*. The hook **132** of each first positioning member **130** is engaged with the corresponding first slot **15**, and the elastic arm **131** abuts against the first positioning surface **13** of the casing **10**.
(13) As shown in FIG. **2** and FIG. **4**, each second positioning member **150** includes an elastic arm **151** and a hook **152**. The elastic arm **151** is connected to the third surface **113**, and the hook **152** is connected to the elastic arm **151**. In detail, the third surface **113** has two grooves **113***a* configured to accommodate the two elastic arms **151** of the two second positioning members **150**. In addition,

the hook **152** of each second positioning member **150** protrudes outward from the corresponding elastic arm **151** and protrudes from the third surface **113**. The hook **152** of each second positioning member **150** is engaged with the corresponding second slot **16**, and the elastic arm **151** abuts against the second positioning surface **14** of the casing **10**.
(14) As shown in FIG. **3** and FIG. **4**, due to the abutting relationship between the elastic arm **131** of each first positioning member **130** and the first positioning surface **13** of the casing **10** and the abutting relationship between the elastic arm **151** of each second positioning member **150** and the second positioning surface **14** of the casing **10**, the force-bearing bracket **110** may not easily shake in the first direction D**1**, so the idle stroke generated when the first button **120***a* or the second button **120***b* is pressed is mitigated.
(15) As shown in FIG. **4**, the hook **132** of each first positioning member **130** has a stopping surface **132***a* abutting against an inner wall surface **15***a* of the corresponding first slot **15**. Similarly, the hook **152** of each second positioning member **150** has a stopping surface **152***a* abutting against an inner wall surface **16***a* of the corresponding second slot **16**. As shown in FIG. **3** and FIG. **4**, by the abutting relationship between the hook **132** of each first positioning member **130** and the corresponding first slot **15** and the abutting relationship between the hook **152** of each second positioning member **150** and the corresponding second slot **16**, the force-bearing bracket **110** may not easily shake in the second direction D**2**, and the idle stroke generated when the first button **120***a* or the second button **120***b* is pressed is mitigated.
(16) As shown in FIG. **3** to FIG. **5**, the casing **10** also has an inner surface **17** located between the first positioning surface **13** and the second positioning surface **14**. Each elastic member **140** abuts against the inner surface **17** and may be in a compressed state to strengthen the engaging relationship between the first positioning member **130** and the casing **10**, and the engaging relationship between the second positioning member **150** and the casing **10** through the elastic restoring force of the elastic member **140**, so that the force-bearing bracket **110** may not easily shake in the second direction D**2**, so the idle stroke generated when the first button **120***a* or the second button **120***b* is pressed is mitigated. As shown in FIG. **1**, the elastic member **140** may be a semicircular elastic arm, a semilunar elastic arm, or an arced elastic arm.
(17) As shown in FIG. **2** and FIG. **4**, in this embodiment, each second positioning member **150** further includes an operating portion **153**, which is connected to the elastic arm **151**, and the hook **152** is located approximately at a junction of the operating portion **153** and the elastic arm **151**. On the other hand, the bracket **110** also has a fourth surface **114** opposite the first surface **111**. The operating portion **153** is located outside the groove **113***a* and protrudes from the fourth surface **114**.
(18) For an operator, the operating portion **153** may be used as a force application point for disengaging the engaging relationship between the second positioning member **150** and the casing **10**, and the operator may first apply force to pull the operating portion **153** of the two second positioning members **150**, so that the two hooks **152** are moved out of the two second slots **16** of the casing **10**. Next, the two hooks **132** of the two first positioning members **130** are moved out of the two first slots **15** of the casing **10**. After completely disengaging the engaging relationship between the bracket **110** and the casing **10**, the operator may disassemble the button module **100** from the casing **10**, so the disassembly and assembly are very convenient.
(19) In particular, although the quantities of the first positioning member **130**, the elastic member **140**, the second positioning member **150**, the first slot **15**, the second slot **16**, the groove **112***a*, and the groove **113***a* have been illustrated as above, the quantities are not limited thereto. In other embodiments, the quantities of the first positioning member **130**, the elastic member **140**, the second positioning member **150**, the first slot **15**, and the second slot **16** may be increased or decreased according to design requirements.
(20) In summary, in the button module of the disclosure, the positioning member and the elastic member are integrated on the bracket, and the bracket may be fixed to the casing through the positioning member, which is not only convenient for assembly, but also less prone to breakage of

the bracket or casing due to assembly errors, so that assembly time can be reduced and assembly yield can be improved. On the other hand, the bracket may abut against the casing through the elastic member. When the user presses the button, the elastic member may feed back to the user a relatively solid operating texture. Moreover, based on the cooperation between the positioning member and the elastic member, the force-bearing bracket may not easily shake, and therefore the idle stroke generated when the button is pressed is mitigated, thereby improving the operating reliability. In addition, the bracket and the casing are fixed by adopting a screwless manner, which helps to save manufacturing costs.
(21) Although the disclosure has been disclosed in the above embodiments, the embodiments are not intended to limit the disclosure. Persons skilled in the art may make some changes and modifications without departing from the spirit and scope of the disclosure. Therefore, the protection scope of the disclosure shall be defined by the appended claims.

Claims

1. A button module, comprising: a bracket, having a first surface and a second surface connected to the first surface; at least one button, disposed on the first surface; at least one first positioning member, comprising an elastic arm and a hook connected to the elastic arm, wherein the elastic arm is connected to the second surface, and the hook is engaged with a casing; and at least one elastic member, protruding from the first surface and abutting against the casing.
2. The button module according to claim 1, wherein the at least one elastic member comprises a semicircular elastic arm, a semilunar elastic arm, or an arced elastic arm.
3. The button module according to claim 1, wherein the at least one first positioning member comprises two first positioning members, and the at least one elastic member is located between the two first positioning members.
4. The button module according to claim 3, wherein the at least one elastic member comprises two elastic members, and the two elastic members are arranged side by side between the two first positioning members.
5. The button module according to claim 1, wherein the second surface has at least one groove, and the elastic arm extends outward from the at least one groove and protrudes from the first surface.
6. The button module according to claim 5, wherein the hook protrudes from an end of the elastic arm and is located outside the at least one groove.
7. The button module according to claim 1, further comprising: at least one second positioning member, wherein the bracket further has a third surface opposite the second surface, the first surface is connected to the second surface and the third surface, and the at least one second positioning member comprises an elastic arm and a hook connected to the elastic arm, wherein the elastic arm of the at least one second positioning member is connected to the third surface, and the hook of the at least one second positioning member is engaged with the casing.
8. The button module according to claim 7, wherein the third surface has at least one groove, the elastic arm of the at least one second positioning member is located in the at least one groove of the third surface, and the hook of the at least one second positioning member protrudes from the third surface.
9. The button module according to claim 7, wherein the casing has a first positioning surface, a second positioning surface opposite the first positioning surface, at least one first slot located on the first positioning surface, and at least one second slot located on the second positioning surface, the second surface of the bracket faces the first positioning surface, the third surface of the bracket faces the second positioning surface, and the hook of the at least one first positioning member is engaged with the at least one first slot, wherein the hook of the at least one first positioning member has a stopping surface abutting against an inner wall surface of the at least one first slot, and the hook of the at least one second positioning member is engaged with the at least one second

slot, wherein the hook of the at least one second positioning member has a stopping surface abutting against an inner wall surface of the at least one second slot.

10. The button module according to claim 7, wherein the at least one second positioning member further comprises an operating portion, the operating portion is connected to the elastic arm of the at least one second positioning member, the bracket further has a fourth surface opposite the first surface, and the operating portion protrudes from the fourth surface.